

FBO ANNOUNCEMENT: PRESOLICITATION NOTICE, NOTICE OF INTENT

ACTION CODE: SPECIAL NOTICE

CLASSIFICATION CODE: 66 – Instruments and Laboratory Equipment

SUBJECT: Gas Cylinders

SOLICITATION NUMBER: NB646030-25-00803

DESCRIPTION:

The National Institute of Standards & Technology (NIST) intends to negotiate on a brand name basis under the authority of FAR 13.105, for Luxfer brand SGS & Non-SGS Aluminum Gas Cylinders.

Brand Name Justification is based on the following:

The Material Measurement Laboratory's Gas Sensing Metrology Group (GSMG) of the Chemical Science Division is committed to supporting the NIST gas Standard Reference Materials (SRMs) and NIST-Traceable Reference Materials (NTRMs) programs. To support the SRM and NTRM programs, GSMG maintains groups of Primary gas Standard Materials (PSMs). Many gas PSMs are depleted, and GSMG is in urgent need of producing replacement PSMs. GSMG requires new, highly stable, non-reactive aluminum gas cylinders for the production of new PSMs for both non-reactive and reactive gas mixtures.

The production of PSMs requires the use of stable, long lifespan gas cylinders that exceed the specifications found in industry. These gas cylinders are required to hold gas mixtures stable for several years without any change in the composition of the gas. Gas mixtures range from non-reactive to reactive and at concentrations that range from low to high. Sources of deviation from long-term stability include impurities and moisture trapped in the cylinder walls during production, reactions with the gas cylinder walls, reactions with the gas cylinder valve, and the presence of any leaks. Particularly for the low concentrations that NIST sometimes uses, even small levels of reactivity will affect the stability of the gas mixture after years of containment.

With proper maintenance typical cylinders provide mechanical stability last for about 10 years, but not implied that typical cylinders maintain chemical stability of the gas mixture for 10 years. NIST requires a cylinder that provides chemical stability of gas mixtures for a minimum of 10 years, as verified by NIST-internal stability measurements. Some cylinders still perform well and the measurements of gas components inside are still stable after over 30 years.

Market research by the Government has identified the Luxfer brand SGS & non-SGS Aluminum gas cylinders, as the only cylinders capable of long-term chemical stability. Five to ten years of testing is required for stability testing. The Luxfer brand aluminum cylinders have continually demonstrated its met requirements for over 30 years. Other branded cylinders have not been tested thus, not enough evidence for the long-term stability to determine if the cylinders can meet requirements.

NAICS code for this procurement will be 332420 – Metal Tank (Heavy Gauge) Manufacturing.

No solicitation package will be issued. This notice of intent is not a request for competitive quotations; however, interested parties who regard themselves as capable of fulfilling these requirements are invited to submit responses to the Government for consideration. Responses received by 9/10/2024 at 10:00 am will be considered by the Government.

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